ATM32x1 Harvesting Meter Application Note

Revision History

Date	Version	Description
July 2, 2020	0.50	Initial version created.
March 9, 2021	0.51	Updated Figure 8 - Harvesting Meter Accuracy vs Input Solar Power, Figure 9 - Harvesting Meter Accuracy vs Input RF Power.
April 14, 2021	0.52	Updated format, no content change.
July 1, 2021	0.53	Updated Hardware Setup
September 30, 2021	0.54	Updated 2 Harvesting Meter Mechanism section, added 5 Software Operation.

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1 Overview

The Atmosic ATM3 series system-on-a-chip (SoC) supports energy harvesting from RF sources or from other modalities such as photovoltaic (solar), thermal (TEG), and mechanical actuation. In order to demonstrate how much energy the ATM3 series can harvest under certain circumstances, a harvesting meter application has been implemented to display harvested energy in real time through UART1 debug print console.

This document includes hardware configuration and programming examples to demonstrate how to enable the harvesting meter using the ATM3 Evaluation Board (EVB) and Software Development Kit (SDK).

NOTE: The harvesting meter application requires ATM32x1 EVB version xxx-03 or newer and Interface Board version 011-02 or newer. Previous versions of the EVB and Interface Board do not support the harvesting meter.

1.1 Related Documentation

Referenced documents are available on the Atmosic Support Website:

- Atmosic SDK Quick Start Guide
- ATM32x1 EVK Energy Harvesting Quick Start Guide
- ATM32x1 Energy Harvesting Application Note
- ATMx2x Reference Manual

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2 Harvesting Meter Mechanism

The M3 runs off the external V_{BATT} supply. In the harvesting application, register overwrites are applied to force all incoming harvesting power to only go to V_{STORE}, which will keep charging the load capacitors (C_{STORE}). During this time the V_{STORE} voltage is periodically digitalized by GADC. By looking at the delta in V_{STORE} over a set amount of time, a measurement of the energy harvested is given by the formula:

$$E = 1/2 * C_{STORE} * V_{STORE}^2$$

$$\Delta E = 1/2 * C_{STORE} * (V_{STORE2}^2 - V_{STORE1}^2)$$

$$P = \Delta E/\Delta t = 1/2 * C_{STORE} * (V_{STORE2}^2 - V_{STORE1}^2)/\Delta t$$

However, when C_{STORE} is at capacity (i.e V_{STORE} is at its maximum voltage) the harvesting meter no longer measures the actual amount of harvested energy. To prevent C_{STORE} from reaching capacity, GPIO13 is connected to V_{STORE} and is asserted low to drain the energy stored on C_{STORE} as it reaches close to its maximum voltage level. The GPIO13 is set back to high impedance mode once V_{STORE} drops below a preset voltage level.

The application software continuously computes the harvested power level and displays it through UART1 debug print in real time.

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3 Hardware Setup

An example of EVB version 013-03-00 and Interface Board version 011-02-00 setup is shown below in Figure 1 and Figure 2, for ATM3201 and ATM3221 respectively.

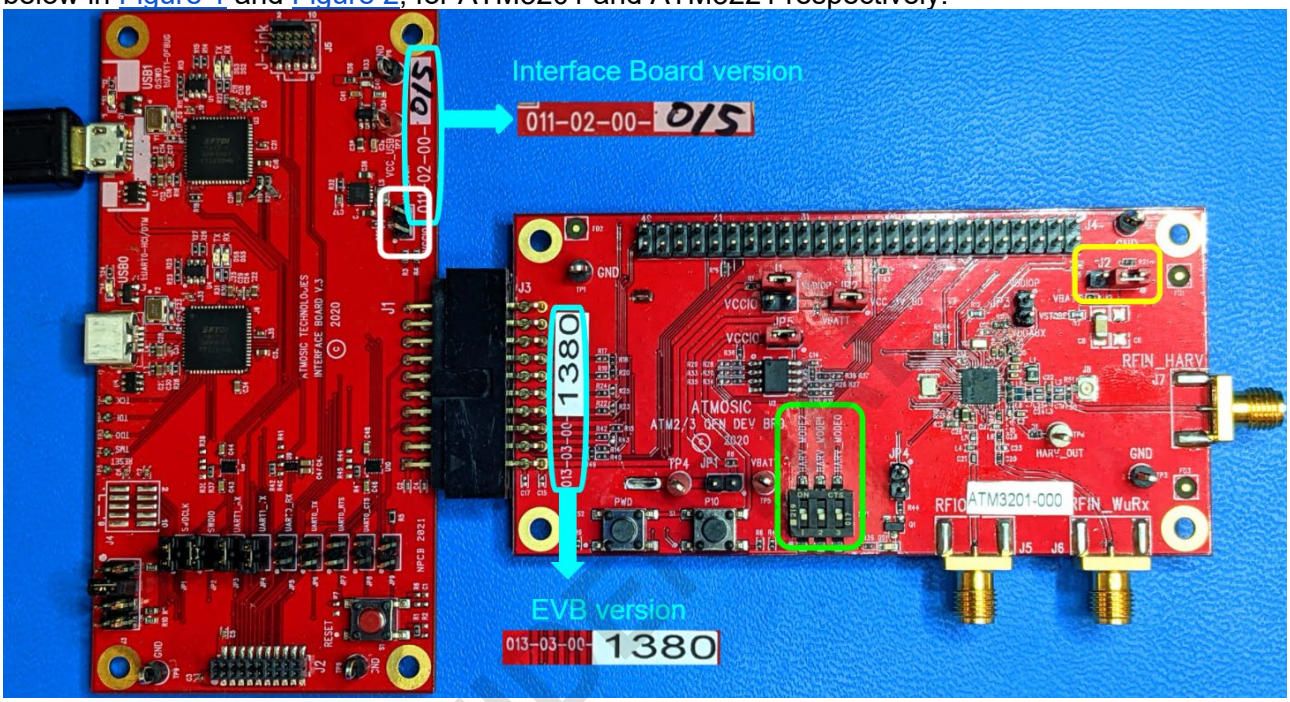


Figure 1 - Harvesting Meter Hardware Setup for ATM3201 EVB

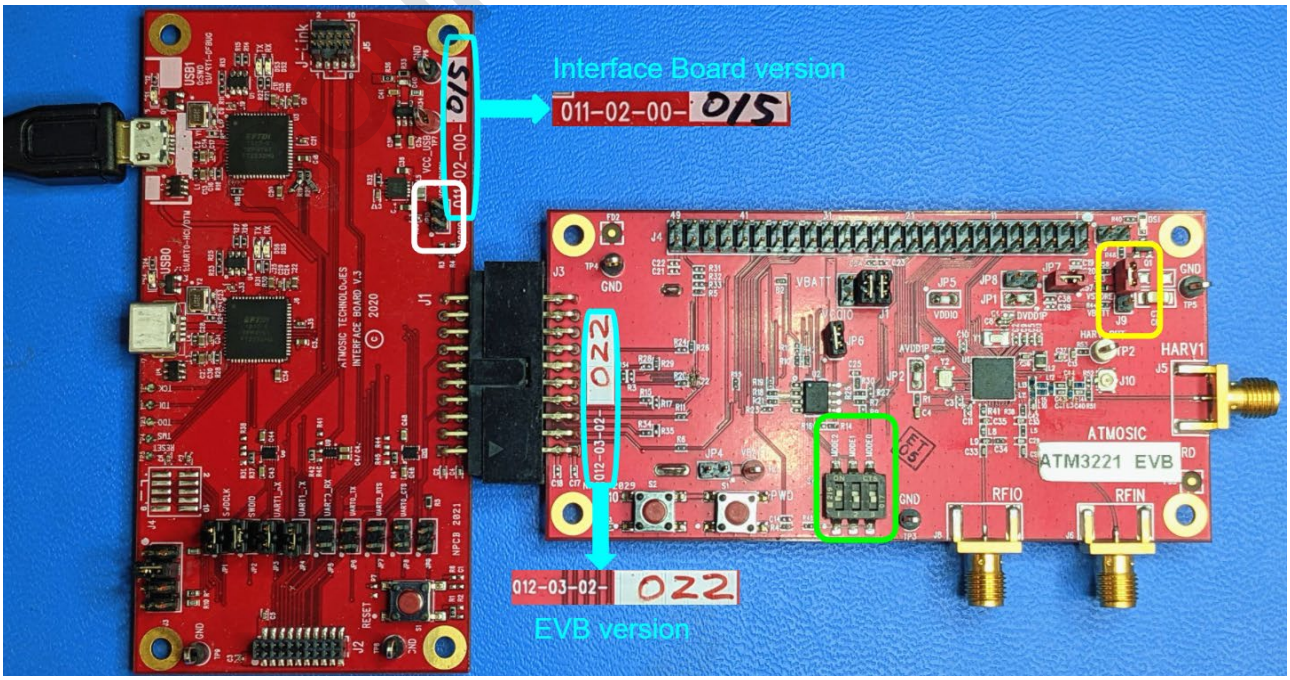


Figure 2 - Harvesting Meter Hardware Setup for ATM3221 EVB

Jumper and DIP switch settings are shown in Table 1.

Board	Jumper / DIP Switch	Position
Interface Board	JP10 (white)	Disconnect
ATM32x1 EVB	J2 on ATM3201 or J9 on ATM3221 (yellow)	Connect 1-2
	SW1 (green)	Figure 3 shows the DIP Switch markings. “ON” on top is the direction for logic high; square dot at bottom left corner is the direction for logic low. Set MODE1 to logic low to enable Harvesting Meter Function. Set MODE0 to logic high for RF harvesting, logic low for non-RF harvesting. MODE2 is currently not programmed.

Table 1 - Harvesting Meter Jumper / DIP Switch Connection

DIP switch setting is demonstrated below in Figure 3. If the device is in operation when the DIP switch position is changed, the EVB must be power cycled before the setting change will take effect.

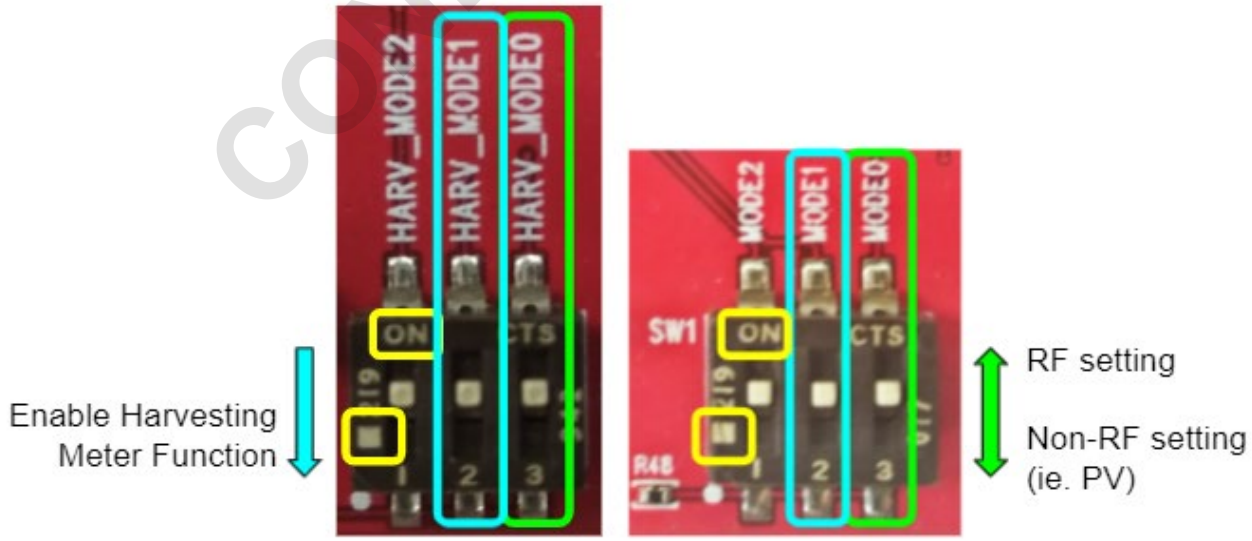


Figure 3 - DIP Switch Setting for ATM3201 (Left) and ATM3221 (Right)

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3.1 Power Cycling the EVB

The Powerdown (PWD) button should ***NOT*** be used for power cycling the board. This function is only supported when an external I/O supply is used.

Instead, the following steps should be taken to power cycle the EVB:

1. Remove the power source by disconnecting the interface board from the EVB
2. Wait at least 10 seconds for power rails to discharge
3. Reconnect the EVB to the interface board

4 Software Setup

The Harvesting Meter application is a compile time option in the BLE_harv_adv example application makefile. Uncomment below lines shown in Figure 4 to enable the Harvesting Meter functionality.

```
# Harv Meter
# Uncomment lines below to enable Harvesting Meter functionality
# Refer to README file for descriptions for various parameter settings.
#
DRIVERS += gadc interrupt sw_event timer
CFLAGS += -DCFG_HARV_METER -DCFG_GADC -DNO_BROWNOUT=1
CFLAGS += -DVSTORE_HWM=2.1 -DVSTORE_LWM=0.65 -DSAMPLE_RATE_CHARGE_IN_MS=10 -DCAP_VAL=220
```

Figure 4 - Compile Option for Harvesting Meter Application

To compile and load the harvesting meter application:

1. In the SDK example BLE_harv_adv folder, uncomment above lines in the makefile
2. Issue “make clean”
3. Issue “make run_all” to compile and load the image to target

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5 Software Operation

Software starts a timer to measure vstore every 100 ms (SAMPLE_RATE_CHARGE_IN_MS). During the charging phase, software computes the amount of harvested energy every 1 second (RECORD_INTERVAL_MSEC) and displays it on the serial console.

Once vstore voltage reaches high water mark 2.5 V (VSTORE_HWM), energy is drained by driving GPIO13 (HARV_METER_GPIO) low. Discharging is expected to be completed in 100 ms (SAMPLE_RATE_DISCHARGE_MS).

On completion of discharge, the charging cycle begins again and repeats the above sequence continuously.

6 Harvesting Meter Performance

Connect EVB and Interface Board as shown above in Figure 5. Then connect the Interface Board USB1 port to a PC. Check COM Port number in Device Manager:

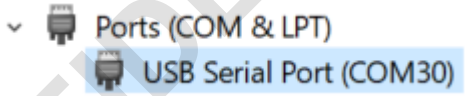


Figure 5 - COM Port Setting in Device Manager

To access the UART1 debug printing via Putty:

1. Select “Serial”
2. Set “Serial line” to the corresponding COM port number shown in the Device Manager
3. Set “Speed” to 115200
4. Click “Open”

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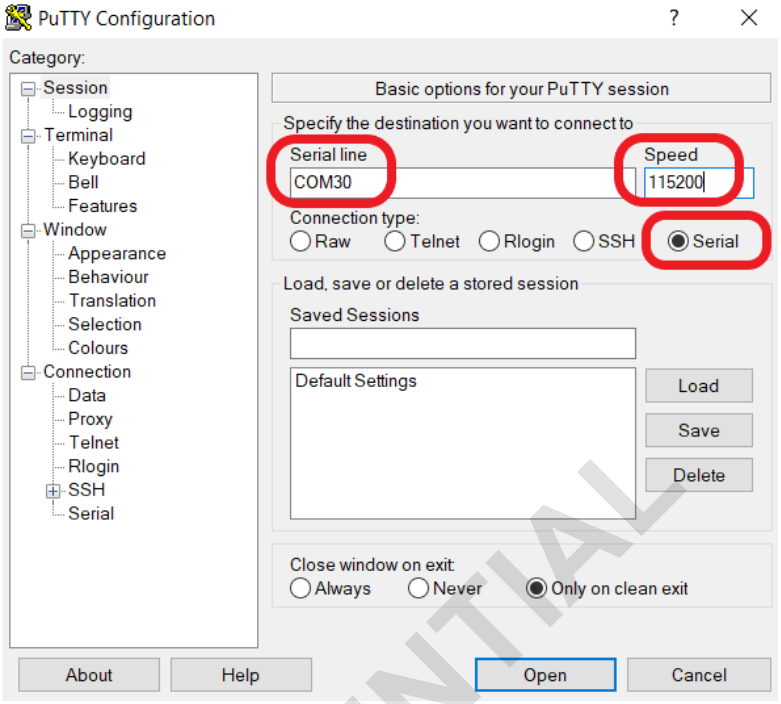


Figure 6 - Putty Setting for UART1 Debug Print

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Figure 7 shows an example of the debug printout of the energy harvesting measurements.

```
@0000002c E3d silicon
5x5 EXT_FLASH: 4e 56 44 53 06 06 27 00 00 00 00 01 02 00 00 00 ...
@000001a3 user_main() done
@00000301 Advertising: ON
@00008ee0 Harvested Energy: 0.046999 mW
@00010edf Harvested Energy: 0.040207 mW
@00018ede Harvested Energy: 0.045179 mW
@00020edd Harvested Energy: 0.046923 mW
@00028edd Harvested Energy: 0.050601 mW
@00030edc Harvested Energy: 0.053176 mW
@00036875 Harvested Energy: 0.053804 mW
@00036900 Harv Meter: Discharging
@00036b02 Harv Meter: Charging
@0003eb79 Harvested Energy: 0.044881 mW
@00046b78 Harvested Energy: 0.043496 mW
@0004eb77 Harvested Energy: 0.040767 mW
```

Figure 7 - Harvesting Meter Debug Print Console Example

Harvested energy is printed out every 1 second. At high harvesting power, information will be printed whenever V_{STORE} reaches the preset maximum voltage level.

“Harv Meter: Discharging” denotes that V_{STORE} reached the preset maximum voltage level, and GPIO13 is asserted low to drain the energy stored on C_{STORE}. Similarly, “Harv Meter: Charging” denotes that V_{STORE} drops below the preset minimum voltage level, and GPIO13 is set back to high impedance mode.

Figure 8 and Figure 9 are examples of overall Harvesting Meter accuracy over various input power levels. Both RF harvesting and non-RF harvesting show +/-10% accuracy. Users should expect some board to board variation. Please refer to the plots below for more details.

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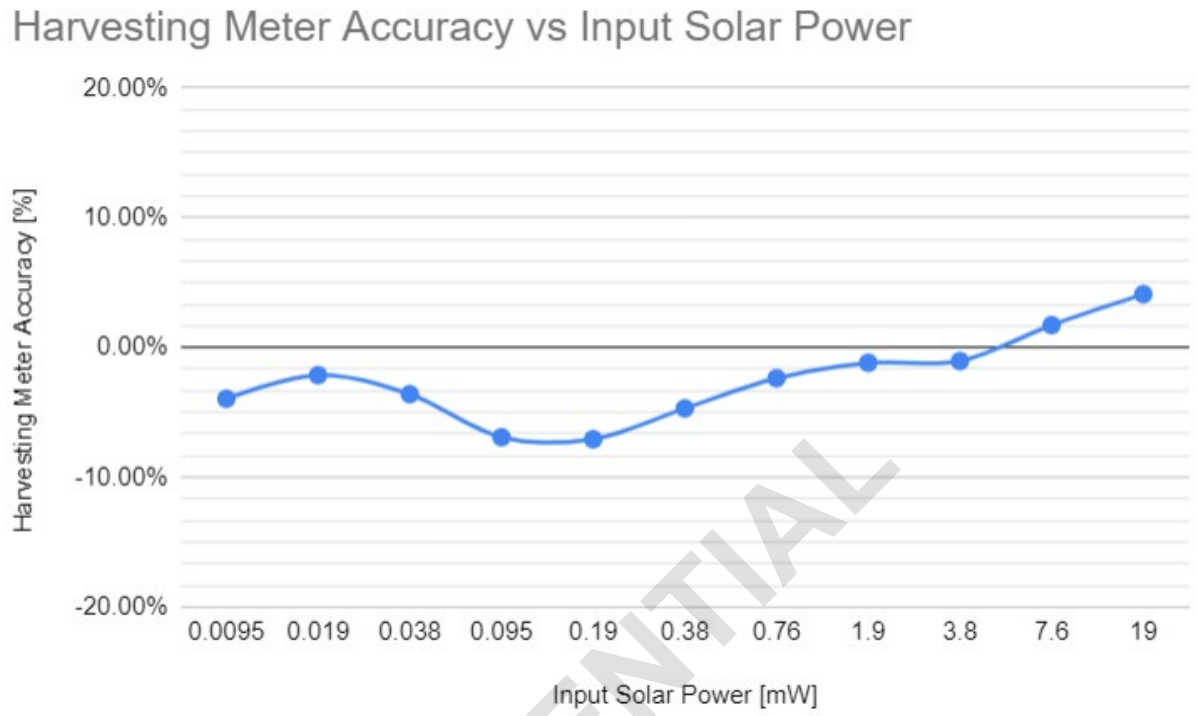


Figure 8 - Harvesting Meter Accuracy vs Input Solar Power

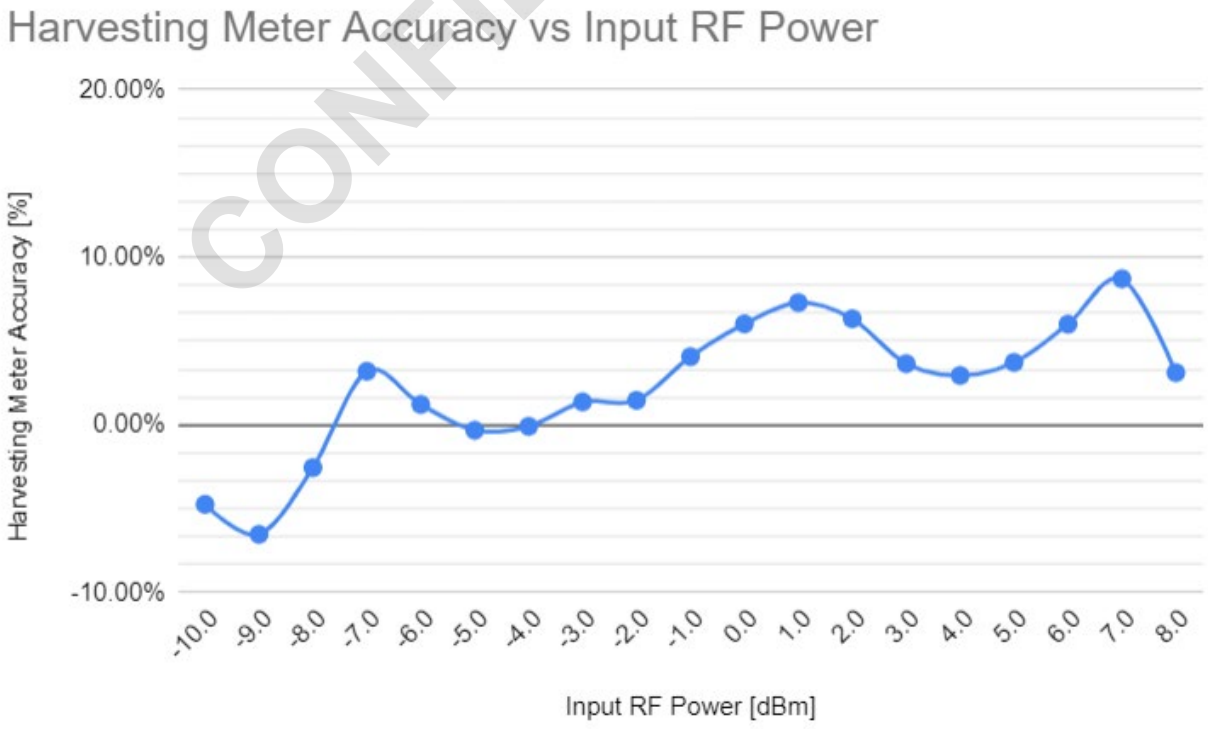


Figure 9 - Harvesting Meter Accuracy vs Input RF Power

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